Muhammad Ayaz

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Researcher specializing in Computer Vision, Multimodal AI, Vision-Language Models (VLMs), and other Foundation Models, with experience in Visual Perception and Deep Representation Learning. Research applications include Medical Imaging, Fire and Smoke Analysis, Disaster Scene Understanding, and Affective Computing. Experienced in developing and evaluating robust, efficient, and deployable Deep Learning systems using Python, PyTorch, and TensorFlow.

EDUCATION

MS, Applied Arts and Engineering *2025 - Present*
Chung-Ang University (CAU), Graduate School of Applied Arts and Engineering *Seoul, South Korea*
- GPA: 4.33/4.5 (98%); current laboratory: Graphics Realization Lab (GRLab).
- Master's Thesis (in progress): "Beyond Scene Recognition: Operational Reasoning for Disaster Response with Vision-Language Models."

BS Computer Science *2019 - 2023*
The University of Agriculture Peshawar (AUP) *Peshawar, Pakistan*
- CGPA: 3.30/4.0 (75.91%).
- Thesis: "Toward Reliable Fire & Smoke Instance Segmentation in Challenging Environments: A Vision-Based Approach."

PROFESSIONAL EXPERIENCE

Research Assistant *Mar 2025 - Present*
Graphics Realization Lab (GRLab), Chung-Ang University *Seoul, South Korea*
- Conduct research in computer vision, image segmentation, multimodal learning, and deep learning under Prof. Sanghyun Seo.
- Develop robust models for fire and smoke understanding, medical vision-language reasoning, and real-world visual perception.

Research Assistant *Sep 2022 - Dec 2024*
Digital Image Processing Laboratory, Islamia College University *Peshawar, Pakistan*
- Contributed to image classification, 2D and 3D object detection, image segmentation, autonomous driving, and video understanding projects.
- Supported dataset preparation, model training, experimental evaluation, and research documentation.

Course Assistant *May 2023 - Dec 2024*
Digital Image Processing Laboratory, Islamia College University *Peshawar, Pakistan*
- Assisted with courses in artificial intelligence, computer vision, machine learning, and deep learning.

Junior Researcher *Dec 2020 - 2022*
National Center for Big Data and Cloud Computing (NCBC) *Peshawar, Pakistan*
- Participated in data collection, annotation, preprocessing, and computer vision research tasks.

SELECTED PROJECTS

ROAD-R Challenge for NeurIPS 2023 *2023*
- Worked on autonomous-driving event understanding using the ROAD-R dataset and logical requirements.

MedVLM: Vision-Language Model for Holistic Medical Applications *2024 - 2025*
- Trained and evaluated multimodal models for medical visual question answering using X-rays, CT scans, and MRIs.
- Collaborated with researchers at MBZUAI and United Arab Emirates University.

Cone Detection for Formula Student Cars
- Developed and fine-tuned object detection models for cone-based vehicle navigation using the FSOCO dataset.

Fire and Smoke Instance Segmentation in Challenging Environments
- Developed vision models for accurate fire and smoke instance segmentation across indoor and outdoor scenes.
- Improved robustness under difficult lighting, occlusion, scale variation, and complex environmental conditions.

Anomaly Detection in Video Sequences
- Combined convolutional feature extraction with recurrent temporal modeling to identify abnormal video events.

SELECTED PUBLICATIONS

1. BoXFire: Geometry-Aware Box-Supervised Fusion for Accurate Fire and Smoke Segmentation. *Advanced Engineering Informatics*, 2026. **First author.**
2. Medical Question Answering: A Comprehensive Multimodal and LLM-Driven Survey. *Computer Methods and Programs in Biomedicine*, 285, 109541, 2026. Co-author.
3. Privacy-Preserving Action Recognition: Taxonomy, Methods, and Privacy-Utility Trade-offs. arXiv:2608.04501, 2026. Co-author.
4. MedVLM: Vision-Language Model for Holistic Medical Applications. *IEEE Consumer Electronics Magazine*, 2025. **First author.**
5. **Box-Supervised Fire and Smoke Segmentation in Real-World Scenarios**. 21st International Conference on Multimedia Information Technology and Applications (MITA), 2025. **Best Paper Award**; press coverage.
6. WVES: A Weighted Ensemble Framework for Student Mental Health Assessment. *2026 International Conference on Technical Education (ICTechED)*.

RESEARCH COLLABORATIONS

Dr. Salman Khan, Research Fellow, Visual Artificial Intelligence Laboratory, Oxford Brookes University, UK
- Collaborated on cone detection, road-crack annotation, fire detection, and introductory 3D object detection using the KITTI dataset.

Dr. Mustaqeem Khan, Assistant Professor, United Arab Emirates University, UAE
- Collaborated on medical vision-language modeling and multimodal violence-detection experiments.

Dr. Alice Othmani, Postdoctoral Researcher, Ecole Normale Superieure de Paris, France
- Collaborated on Medical Question Answering: A Comprehensive Multimodal and LLM-Driven Survey.
- Research interests include pattern recognition, image and mesh analysis, contour processing, and shape-feature extraction (homepage).

ACHIEVEMENTS & SCHOLARSHIPS

- **Best Paper Award**, 21st MITA International Conference on Multimedia Information Technology and Applications, 2025 (press coverage).
- **Young Scientist Scholarship**, Chung-Ang University, South Korea, 2025 - Present.
- Fully funded Higher Education Commission scholarship, Pakistan, 2019 - 2023.
- Certificate in Artificial Intelligence, specializing in machine learning and deep learning, awarded by Pakistan's National Vocational and Technical Training Commission.

TECHNICAL SKILLS

Programming Languages	Python, C++, C#, SQL
Python Libraries	OpenCV, Scikit-learn, Scikit-image, NumPy, Matplotlib
Deep Learning Frameworks	PyTorch, TensorFlow, Keras
Research Areas	Computer vision, multimodal AI, vision-language models, segmentation, medical AI

REFERENCES

Dr. Salman Khan
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